

试卷编号：____；注意：1 答案全部手写在 A4 纸上；2 答案拍照或扫描后合并成一个 PDF 文件，注意页面顺序、清晰度；3 文件命名方式“教师点名册上序号+姓名+班级”，5 班发到 sd xujie@163.com, 6 班发到 penghui 0626@163.com。

广东工业大学考试试卷

2019 年—2020 学年第 2 学期

课程名称：____ 自动控制原理 ____ 学分 3 ____ 试卷满分 100 分

考试形式：开卷

考试时间：2020 年 7 月 6 日（第 19 周 星期 一）

题 号	一	二	三	四	五	六	七	八	九	十	总分
评卷得分											
评卷签名											
复核得分											
复核签名											

1. For system shown in Fig. 1, determine the relationship between input voltage and output voltage, express this relationship in the form of a transfer function. (10 marks)

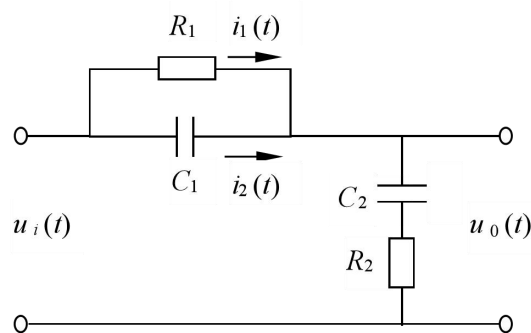


Fig. 1

2. For system shown in Fig. 2, determine the closed-loop transfer function $\frac{C(s)}{R(s)}$. (10 marks)

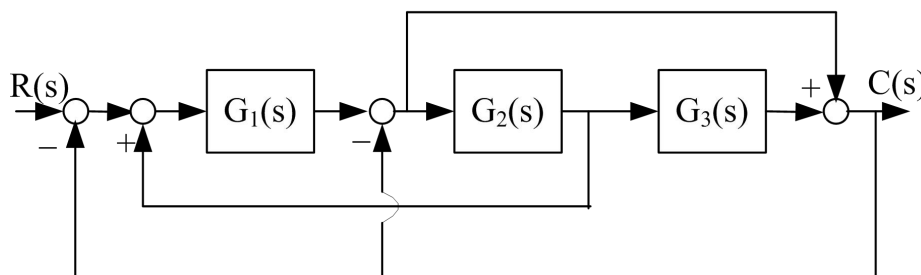


Fig. 2

3. The system block is shown as in Fig. 3 with $K=10$,

(a) if $\tau = 0$, determine the system damping ratio ξ , natural frequency ω_n and the steady-error under unit ramp input; (8 marks)

(b) if system damping ratio $\xi = 0.6$, determine τ and the steady-error under unit ramp input. (6 marks)

(c) if $\xi = 0.6$, the steady-error $e_{ss} = 0.2$ under unit ramp input, determine τ and K . (6 marks)

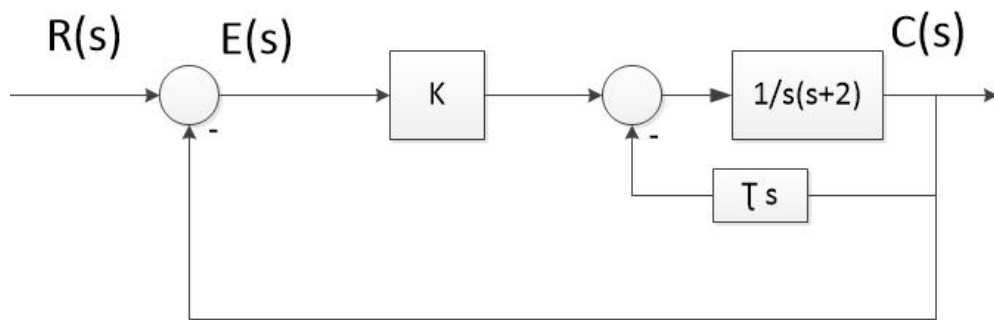


Fig. 3.

4. Draw the root locus for the unit negative feedback system with open-loop transfer function

$GH(s) = \frac{K(s+1)}{(s-1)^2}$ (10 marks), determine the range of K that the closed-loop system is under

damped oscillation (衰减振荡) (4 marks), and the closed-loop transfer function that the system has a pole equals -2 (6 marks).

5. A system has the following open-loop transfer function $G_0(s) = \frac{8}{s(s+2)}$, design a

compensation element that enforces the closed-loop system satisfies

(1) steady-error is 0.05 under $r(t)=t$

(2) phase margin $\gamma \geq 45^\circ$ and the 开环截止频率 $\omega_c \geq 10\text{rad/s}$ (15 marks), and plot the bode diagram of the compensated system. (5 marks)

6. For system shown in Fig. 4,

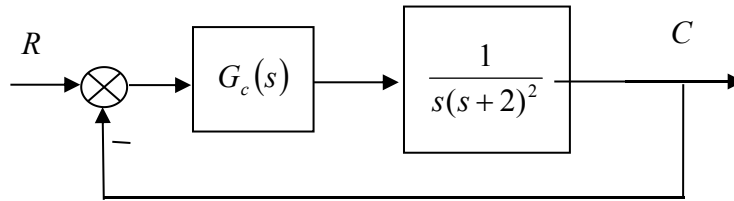


Fig. 4

(a) if $G_c(s)$ is a PID controller, choose suitable K_p , T_i , T_d by Ziegler and Nichols second method and write the transfer function of the controller; (10 marks)

TABLE 21.2. Ziegler-Nichols parameter settings, method 2

Control	K_p	T_i	T_d
P	$0.5K'_p$	∞	0
PI	$0.45K'_p$	$0.833P'$	0
PID	$0.6K'_p$	$0.5P'$	$0.125P'$

(b) sketch the Nyquist diagram of plant $\frac{1}{s(s+2)^2}$, and determine stability of the plant by using the Nyquist stability criterion. (10 marks)